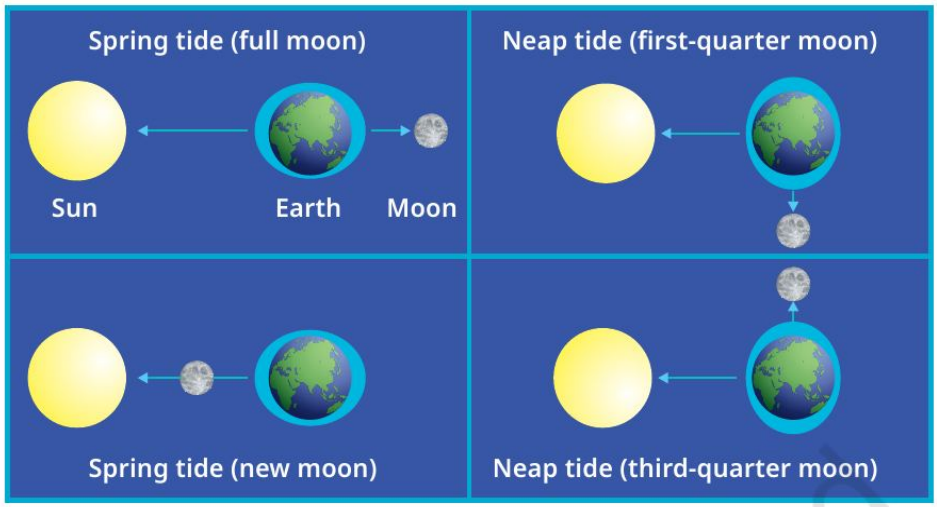


(a) *(b)*

Fig. 1.5 (a) Spring tide (b) Neap tide (Schematic diagram of tidal forcing showing idealised bulges)

Spring Tides and Neap Tides

You may wonder what causes this regular rise and fall of ocean water. To understand this, let us explore how tides are formed.

- **Spring tides:** Observe Fig. 1.5(a), where the Sun, the Earth, and the Moon are aligned in a straight line. Spring tides occur during this alignment, which happens on full moon and new moon days. At this time, the combined gravitational pull of the Sun and the Moon causes high tides of unusual heights and low tides that are unusually lower. The figure also shows the bulging of water on opposite sides of the Earth due to this strong gravitational pull, leading to the formation of spring tides.
- **Neap tides:** When the Moon is in its first- and third-quarter phases, as shown in Fig. 1.5(b), the gravitational pull of the

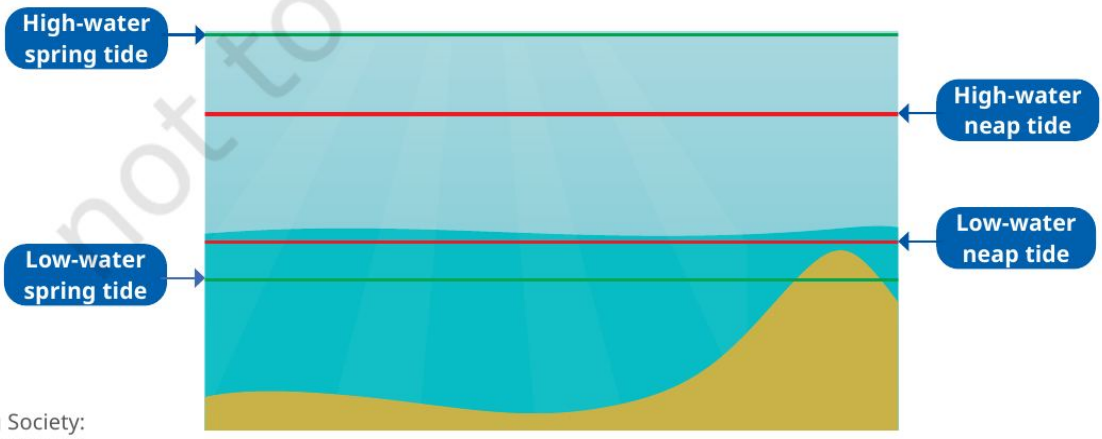


Fig. 1.6. Levels of spring and neap tides

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